MATH 128a: Homework #4

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Due by: **Wednesday, October 1, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Problems from textbook Section 2.1

Solve problems 19(17) from Section 2.1 of the textbook.
If the problem has a different number in edition 9 of the textbook, it is in parentheses.

Answer of Problem 1

Solution. First we observe that for each $n > 1$, we have $p_n - p_{n-1} = \sum_{k=1}^{n}(1/k) - \sum_{k=1}^{n-1}(1/k) = \frac{1}{n}$.
So $\lim_{n\to\infty}(p_n - p_{n-1}) = \lim_{n\to\infty}\frac{1}{n} = 0$.
To see that the sequence diverges, we consider the following sum:

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \cdots$$

Each group of repeated terms sums to $\frac{1}{2}$, and there are in infinite number of groups, so the series diverges. Compare this to our sequence $\sum_{k=1}^{n}(1/k)$, which looks like:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \frac{1}{14} + \frac{1}{15} + \frac{1}{16} + \cdots$$

Each term in this sequence is at least as big as the corresponding term in the previous sequence, which we argued was divergent. By the comparison test, $\sum_{k=1}^{\infty}(1/k)$ diverges too.
Alternatively, the integral test from calculus shows that $\sum_{k=1}^{n}\frac{1}{k} \geq \int_1^{n+1}\frac{1}{x}\,dx = \ln(n+1)$, which grows without bound as $n \to \infty$.

Problem 2: Problems from textbook Section 2.2

Solve problems 1d, 10(8), 13ab(11ab), 21(17), 22(18) from Section 2.2 of the textbook.
If the problem has a different number in edition 9 of the textbook, it is in parentheses.
13a(11a): use the starting interval [a,b]=[0,1].
13b(11b): use the starting interval [a,b]=[2.5,3].
22(18): the 9th edition book has two typos:
Part (a) should refer to Theorem 2.3.
Part (b) should refer to Theorem 2.4.

Answer of Problem 2

Exercise 2.2.1. Use algebraic manipulation to show that each of the following functions has a fixed point at p precisely when $f(p) = 0$, where $f(x) = x^4 + 2x^2 - x - 3$.

d. $g_4(x) = \frac{3x^4+2x^2+3}{4x^3+4x-1}$

Solution. The following equations are equivalent (provided $4x^3 + 4x - 1 \neq 0$):

$$\begin{aligned} x &= \frac{3x^4 + 2x^2 + 3}{4x^3 + 4x - 1}, \\ x(4x^3 + 4x - 1) &= 3x^4 + 2x^2 + 3, \\ 4x^4 + 4x^2 - x &= 3x^4 + 2x^2 + 3, \\ x^4 + 2x^2 - x - 3 &= 0. \end{aligned}$$

Let $h(x) = 4x^3+4x-1$ and $k(x) = 3x^4+2x^2+3$. We claim first that $h(x)$ and $k(x)$ do not share any common factors. If they did, we would have to check that simplifying the formula for $k(x)/h(x)$, which would cause $g_4(x)$ to become well-defined at the common factors, does not lead to additional fixed points $x = g_4(x)$ satisfying $h(x) = 0$. But if they share a common factor, say $(x - x_0)$, then $3xh(x) - 4k(x) = 4x^2 - 3x - 12$ is also divisible by $(x - x_0)$. The roots of this quadratic equation are $x_1 = (3 - \sqrt{201})/8$ and $x_2 = (3 + \sqrt{201})/8$. Since $h(x_1) = -17.4986 \neq 0$ and $h(x_2) = 47.1861 \neq 0$, assuming h and k share a common factor leads to a contradiction. This shows that the fixed points of $x = g_4(x)$ are zeros of $f(x)$. Conversely, if $f(x) = 0$, we claim that $h(x) \neq 0$ so that $x = g_4(x)$. But if $f(x)$ and $h(x)$ shared a common factor $(x - x_0)$, then $xh(x) - 4f(x) = 4x^2 - 3x - 12$ would also be divisibly by $(x - x_0)$. We already showed that $h(x)$ does not share a common factor with this quadratic equation.

Exercise 2.2.10(8). Use Theorem 2.3 to show that $g(x) = 2^{-x}$ has a unique fixed point on $[\frac{1}{3}, 1]$. Use fixed-point iteration to find an approximation to the fixed point accurate to within 10^{-4}. Use Corollary 2.5 to estimate the number of iterations required to achieve 10^{-4} accuracy, and compare this theoretical estimate to the number actually needed.

Solution. To demonstrate the first part of this question using Theorem 2.3 we need to show: (a) $g \in C[\frac{1}{3}, 1]$; (b) $g(x) \in [\frac{1}{3}, 1]$ for all $x \in [\frac{1}{3}, 1]$; (c) g' exists in $(\frac{1}{3}, 1)$; (d) there is a constant $k < 1$ such that $|g'(x)| \leq k$ for all $x \in [\frac{1}{3}, 1]$. Since g is infinitely differentiable on $\mathbb{R}$ this takes care of (a) and (c). For (b) we have $g(\frac{1}{3}) = 2^{-1/3} \approx 0.79$ and $g(1) = 1/2$ both of which are in the interval $[1/3, 1]$. Since g is strictly decreasing on all of $\mathbb{R}$, we get that $g(x) \in [\frac{1}{3}, 1]$ for all $x \in [\frac{1}{3}, 1]$. Finally, for (d), we have $g'(x) = -\ln 2 \cdot 2^{-x}$ and $g'(\frac{1}{3}) \approx -0.55$ and $g'(1) \approx -0.35$. Since g' is strictly increasing on all of $\mathbb{R}$, we deduce that $|g'(x)|$ is bounded by the constant $0.56 < 1$ for all $x \in [\frac{1}{3}, 1]$. For the second part of the question, we apply fixed point iteration starting with $p_0 = 0.66$ using a calculator (or Matlab). The sequence we get is

$$\langle p_n \rangle = \langle .66,\ 0.632878,\ 0.644889,\ 0.639542,\ 0.641917,\ 0.640861,\ 0.64133,\ 0.641122,\ 0.641214,\ \dots \rangle .$$

Computing a few more terms shows that p_8 is correct to 4 decimal places.
The first part of Corollary 2.5 tells us that $|p_n - p| \leq 0.56^n \max\{0.66 - \frac{1}{3}, 1 - 0.66\} = (0.56^n)(0.34)$. Solving $10^{-4} = (0.56^n)(0.34)$ for n gives $n = \ln\left(\frac{0.0001}{0.34}\right) / \ln 0.56 \approx 14.02$, showing that it will take at

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most 15 steps to converge. The second part of Corollary 2.5 tells us that $|p_n - p| \leq \frac{0.56^n}{1-0.56}|0.632878 - 0.66| \approx (0.56^n)(0.0616409)$. Solving $10^{-4} = (0.56^n)(0.0616409)$ gives $n = \ln\left(\frac{0.0001}{0.0616409}\right) / \ln(0.56) \approx 11.08$. This tells us that it takes at most 12 iterations to reach the desired tolerance. The actual implementation performs better than this since it takes only 8 iterations.

Exercise 2.2.13(11). For each of the following equations, determine an interval $[a, b]$ on which fixed-point iteration will converge. Estimate the number of iterations necessary to obtain approximations accurate to within 10^{-5}, and perform the calculations.

a. $g(x) = \frac{2 - e^x + x^2}{3}$

Solution: Plotting $g(x)$ on $[-3, 3]$ shows that $g(x)$ is decreasing and crosses $y = x$ near $x = 0.25$. Thus we try the interval $[a, b] = [0, 1]$. The extrema of $g(x)$ occur at an endpoint since $g'(x)$ is never zero. (Proof: $g'(x) = \frac{2x - e^x}{3}$ goes to $-\infty$ when $x \to \pm\infty$ and has a maximum at $x = \ln 2$, where $g''(x) = \frac{2x - e^x}{3} = 0$. The maximum value of $g'(x)$ is therefore $\frac{2\ln 2 - 2}{3} = -0.20457$.) Since $g(0) = 1/3 \in [0, 1]$ and $g(1) = \frac{3-e}{3} \in [0, 1]$, g maps $[0, 1]$ to itself. It is also continuous since x^2 and e^x are continuous. The maximum value of $|g'(x)|$ occurs at 0, $\ln 2$ or 1. Evaluating $g'(x)$ at these points shows that $|g'(x)| \leq |g'(0)| = 1/3$. Therefore, there is a unique fixed point in $[0, 1]$. If we start the fixed-point iteration in the center of the interval, $p_0 = 1/2$, then we have the bound $|p_n - p| \leq (1/3)^n(1/2)$. Setting this to 10^{-5} and solving for n gives $n = \ln(2 \times 10^{-5}) / \ln(1/3) = 9.84$, so we expect convergence to within 10^{-5} in at most 10 iterations. Since $|p_n - p| \leq \frac{k}{1-k}|p_n - p_{n-1}|$, and $\frac{k}{1-k} = 1/2$, we are guaranteed that the true error is less than 10^{-5} if the algorithm is run with the termination condition $|p_n - p_{n-1}| \leq TOL = 2 \times 10^{-5}$. Doing this in matlab gives $p_9 = 0.2575318$.

b. $g(x) = \frac{5}{x^2} + 2$

Solution: Plotting $g(x)$ on $[1, 4]$ shows that $g(x)$ is decreasing and crosses $y = x$ near $x = 2.7$. The slope is $g'(x) = -10/x^3$, which drops below 1 in magnitude at $x = 10^{1/3} = 2.1544$. Thus, we need a between 2.1544 and 2.7 in order for $|g'(x)|$ to be smaller than 1 and for $[a, b]$ to contain the fixed point near 2.7. We try $a = 2.5$, $b = 3$. Since g is decreasing on $[2.5, 3]$, its extrema occur at the endpoints, $g(2.5) = 2.8 \in [2.5, 3]$ and $g(3) = 23/9 = 2.5556 \in [2.5, 3]$. So g maps $[2.5, 3]$ to itself. Since $|g'(x)| = 10/x^3$ is decreasing, the maximum value of $|g'(x)|$ on $[2.5, 3]$ is $|g'(2.5)| = 0.64 = k < 1$. Since g is continuous on $[2.5, 3]$ (as the denominator is not zero on that interval), there is a unique fixed point for g on $[2.5, 3]$. Using $p_0 = 2.75$ as a starting guess, Corollary 2.5 gives the bound $|p_n - p| \leq (0.64)^n(0.25)$. Setting this to 10^{-5} and solving for n gives $n = \ln(4 \times 10^{-5}) / \ln 0.64 = 22.7$, so the algorithm is guaranteed to converge to this tolerance in 23 iterations. Setting $TOL = \frac{1-k}{k}10^{-5} = \frac{.36}{.64}10^{-5} = 5.625 \times 10^{-6}$, and running matlab with stopping criterion $|p_n - p_{n-1}| \leq TOL$, we obtain $p_{16} = 2.690649$.

Exercise 2.2.21(17). Find a function g defined on $[0, 1]$ that satisfies none of the hypotheses of Theorem 2.3 but still has a unique fixed point on $[0, 1]$.

Solution. Well, for example, take the function $g(x) = \begin{cases} x^2 & 0 \leq x \leq 1/2, \\ -2x & 1/2 < x \leq 1. \end{cases}$

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We see that g is not continuous on $[0,1]$; $g(x)$ takes some values outside $[0,1]$ for some $x \in [0,1]$; g is not differentiable on all of (a,b); and in some of the places where it is differentiable, the derivative is not bounded by a constant less than 1. Yet $g(0) = 0$ is still a unique fixed point for this function in $[0,1]$.

Exercise 2.2.22(18). (a) Show that Theorem 2.3 is true if the inequality $|g'(x)| \leq k$ is replaced by $g'(x) \leq k$, for all $x \in (a,b)$. [*Hint:* Only uniqueness is in question.]

Solution. Let $h(x) = g(x) - x$ and assume $h(x)$ has two zeros, say at $x = p$ and $x = q$, with $a \leq p < q \leq b$. By Rolle's theorem, there is a $\xi \in (p,q)$ such that $h'(\xi) = 0$. But $h'(x) = g'(x) - 1 \leq k - 1 < 0$ for all $x \in [a,b]$, so no such ξ can exist. Thus, $h(x)$ can have only one zero and the fixed point is unique.

(b) Show that Theorem 2.4 may not hold if inequality $|g'(x)| \leq k$ is replaced by $g'(x) \leq k$. [*Hint:* Show that $g(x) = 1 - x^2$, for x in $[0,1]$, provides a conter-example.]

Solution. Following the hint, let $g(x) = 1 - x^2$ for $x \in [0,1]$. We see that $g(x)$ is decreasing on $[0,1]$ and $g(0) = 1$, $g(1) = 0$; thus, g maps $[0,1]$ to itself. Since $g'(x) = -2x \leq 0 = k < 1$, part (a) implies that g has a unique fixed point in $[0,1]$. Explicitly the fixed point is the root of $x^2 + x - 1 = 0$ that lies in $[0,1]$, namely $p = (-1+\sqrt{5})/2$. But fixed point iteration starting with $p_0 = 0$ or $p_0 = 1$ causes $p_{n+1} = 1 - p_n^2$ to cycle back and forth between 0 and 1, so it does not converge to p. In fact, for any $p_0 \neq p$, $p_{n+2} = 1 - p_{n+1}^2 = 1 - (1 - p_n^2)^2 = p_n^2(2 - p_n^2) = p_n + p_n(1 - p_n)(p_n^2 + p_n - 1)$. We note that $p_n^2 + p_n - 1$ is positive if $p_n > p$ and negative if $p_n \in [0, p)$. So $p_{n+2} > p_n$ if $p_n \in (p, 1)$ and $p_{n+2} < p_n$ if $p_n \in (0, p)$. Moreover, $p_{n+1} - p = 1 - p_n^2 - p = p^2 - p_n^2 = -(p_n - p)(p + p_n)$ changes sign each iteration. So p_n jumps from one side of p to the other, getting farther from p every two steps. Thus, fixed point iteration does not converge to p for any starting value other than p itself.

Problem 3: Consistent fixed point formulations

The nonlinear equation

$$f(x) := xe^x - 1 = 0 \tag{1}$$

has the solution

$$x^* = 0.56714329040978\ldots$$

Consistent fixed point formulations of $f(x) = 0$ satisfy $G(x) = x$.
Consider the following consistent fixed point formulations of (1):

$$\begin{aligned} G_1(x) &= e^{-x}, \\ G_2(x) &= \frac{1+x}{1+e^x}, \\ G_3(x) &= x + 1 - xe^x. \end{aligned}$$

Compute 10 iterations for each of these formulations using the nonlinear fixed point iteration starting from value $x_0 = 0.5$. Measure the error to the fixed point x^*. What do you observe? Which reformulations converge and which do not converge? Explain why some of them converge faster/slower or why they don't converge.
You may use the Matlab scripts and functions available on the bcourses page.

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Answer of Problem 3

Listing 1: Computes the stuff asked for in the problem

```
% script to test nonlinear iteration for different fixed point formulations
% 2017/04/26

phi1 = @(x)(exp(-x));
phi2 = @(x)((1+x)./(1+exp(x)));
phi3 = @(x)(x + 1 - x.*exp(x));
maxit = 10;
xroot = 0.56714329040978;
phi1p = @(x)(-exp(-x));
phi2p = @(x)(1./(1+exp(x))-(1+x).*exp(x)./(1+exp(x)).^2);
phi3p = @(x)(1- x.*exp(x)- exp(x));
der1 = phi1p(xroot)
der2 = phi2p(xroot)
der3 = phi3p(xroot)
tol = eps/5;
x0 = 0.5;

[fixed1,noit1,xvec1] = nonlinearIteration(phi1,x0,tol,maxit);
[fixed2,noit2,xvec2] = nonlinearIteration(phi2,x0,tol,maxit);
[fixed3,noit3,xvec3] = nonlinearIteration(phi3,x0,tol,maxit);

err1 = abs(xvec1 - xroot)/xroot;
err2 = abs(xvec2 - xroot)/xroot;
err3 = abs(xvec3 - xroot)/xroot;

figure;
semilogy(1:length(xvec1),err1,'x',1:length(xvec2),err2, ...
    'v',1:length(xvec3),err3,'*');
legend('\phi_1','\phi_2','\phi_3');
```

Listing 2: Iteration Method they got that on the canvas page

```
function [fixed,noit,xvec] = nonlinearIteration(phi,x0,tol,maxit)

    if nargin < 4
        maxit = 10000;
        if nargin < 3
            tol = 10^(-12);
            if nargin < 2
                x0 = 0.5;
                if nargin <1
                    phi = @(x)((1+x)./(1+exp(x)));
                end
            end
        end
    end
```

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```
if nargout > 2
    iterate = 1;
    xvec = x0;
end
xnew = phi(x0);
noit = 1;
if iterate
    xvec = [xvec,xnew];
end

while (noit < maxit && abs(xnew-x0)>= tol)
    x0 = xnew;
    xnew = phi(x0);
    noit = noit + 1;
    if iterate
        xvec = [xvec,xnew];
    end
end

fixed = xnew;
```

We notice that the first and the second fixed point iterations converge, the second one extremely fast, while the third does not seem to converge. We inspect the derivatives of the three functions G_i, $i = 1, 2, 3$:

$$\begin{aligned} G_1'(x) &= -\exp(-x), \\ G_2'(x) &= \frac{1}{1+\exp(x)} - \frac{(1+x)\exp(x)}{(1+\exp(x))^2}, \\ G_3'(x) &= 1 - x\exp(x) - \exp(x). \end{aligned}$$

Evaluating each of these at the fixed point yields:

$$\begin{aligned} G_1'(x^*) &= -0.567143290409786, \\ G_2'(x^*) &= 1.387778780781446 \cdot 10^{-15}, \\ G_3'(x^*) &= -1.763222834351879 \end{aligned}$$

so $|G_i'(x^*)| < 1$ for $i = 1, 2$ and $|G_3'(x^*)| > 1$, and one can check that these statements are also true in a neighborhood of the fixed points, indicating convergence for the first two, while the conditions of the Theorem from class are not satisfied for the third. For the second iteration, the derivative is very small, indicating a very fast rate of convergence.

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